

Distinct Values Subarrays II

Time limit: 1.00 s

Memory limit: 512 MB

Given an array of n integers, your task is to calculate the number of subarrays that have at most k distinct values.

Input

The first input line has two integers n and k .

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print one integer: the number of subarrays.

Constraints

- $1 \leq k \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input	Output
5 2 1 2 3 1 1	10